Hints: Number theory – 8

Class work

The hints to each problem are cumulative. Read one, try again, and open the next only if you are still stuck.

Problem 1

Hint 1. First inequality: α lies between $\frac{p_k}{q_k}$ and $\frac{p_{k+1}}{q_{k+1}}$, and these differ by $\frac{1}{q_k q_{k+1}}$. For Legendre's criterion: if p/q is not a convergent, choose k with $q_k \leqslant q < q_{k+1}$. Show that $|q\alpha - p| \geqslant |q_k\alpha - p_k|$ (best approximation property), and then estimate $\frac{1}{qq_k} \leqslant \left|\frac{p}{q} - \frac{p_k}{q_k}\right|$ to get a contradiction with $|\alpha - \frac{p}{q}| < \frac{1}{2q^2}$.

Hint 2. For $x^2 - 37y^2 = n$ with $0 < |n| < \sqrt{37}$: dividing a solution by $g = \gcd(x, y)$ gives a primitive solution with value n/g^2, which comes from a convergent of $\sqrt{37} = [6; \overline{12}]$. Since the period is 1, every $Q_k = 1$, so convergents give only ± 1 ($6^2 - 37 = -1$, $73^2 - 37 \cdot 12^2 = 1$). Hence $n \in \{\pm 1, \pm 4\}$.

Hint 3. In particular $x^2 - 37y^2 = 3$ has no integer solution. Local solvability: modulo 3^k take $y = 1$ and solve $x^2 \equiv 40$ (Hensel, $40 \equiv 1 \pmod 3$). Modulo 37^k, use $\left(\frac{3}{37}\right) = 1$. Modulo 2^k take $x = 4$; then $y^2 \equiv 13 \cdot 37^{-1} \equiv 1 \pmod 8$ lifts.

Hint 4. For primes $p \nmid 2 \cdot 3 \cdot 37$, the conic has a solution mod p by counting values of x^2 and of $3 + 37y^2$, and it lifts by Hensel. Combine all prime powers by CRT.

Problem 2

Hint 1. Suppose $k = \frac{a^2+b^2}{ab+1}$ is not a square. Among all pairs with this k, take one with $a \geqslant b$ and $a + b$ minimal. View $a^2 - kb\,a + b^2 - k = 0$ as a quadratic in a.

Hint 2. The other root is $a' = kb - a = \frac{b^2-k}{a}$, which is an integer. Show $a' \geqslant 0$: if $a' < 0$, then $a'^2 - kba' + b^2 - k \geqslant a'^2 + k + b^2 - k > 0$. Also $a' \neq 0$, since otherwise $k = b^2$.

Hint 3. Then $a' = \frac{b^2-k}{a} < \frac{b^2}{a} \leqslant b \leqslant a$, so (a', b) is a smaller pair with the same k, contradicting minimality.

Problem 3

Hint 1. Fix c and take a solution with $a \geqslant b$ and $a + b$ minimal. Vieta jumping in a: the other root is $a' = bc - a = \frac{b^2+1}{a}$, a positive integer.

Hint 2. If $a > b$, then $ab \geqslant b^2 + b \geqslant b^2 + 1$, so $a' \leqslant b < a$ and (a', b) is a smaller solution, a contradiction. Hence $a = b$, and $a^2c = 2a^2 + 1$ gives $a^2 \mid 1$. So $a = b = 1$ and $c = 3$.

Hint 3. Running the jumps backwards from $(1, 1)$ with $c = 3$ gives all pairs: consecutive terms of $1, 1, 2, 5, 13, 34, 89, \ldots$ ($x_{k+1} = 3x_k - x_{k-1}$, the odd-indexed Fibonacci numbers), in either order.

Problem 4

Hint 1. If $q = 0$, r is rational, and $\begin{pmatrix} 1 + rt & -r^2t \\ t & 1 - rt \end{pmatrix}$ works for a suitable integer $t \neq 0$ that clears denominators. Otherwise r is a root of $ux^2 + vx + w$ with integer coefficients and discriminant $v^2 - 4uw = 7\Delta^2$, where Δ is an integer (e.g. $\Delta = 2uq$).

Hint 2. A matrix fixing r has $cr^2 + (d - a)r - b = 0$, so try $c = ut$, $d - a = vt$, $b = -wt$. Writing $a + d = s$, the condition $ad - bc = 1$ becomes $s^2 - 7\Delta^2 t^2 = 4$.

Hint 3. Solve this Pell-type equation with $t \neq 0$. Take $k + l\sqrt{7} = (8 + 3\sqrt{7})^n$ with $\Delta \mid l$, which is possible because the powers are periodic modulo Δ. Then set $s = 2k$, $t = 2l/\Delta$, and check that $a = \frac{s-vt}{2}$, $d = \frac{s+vt}{2}$ are integers.

Problem 5

Hint 1. Put $P = x_1 \cdots x_{n-1}$. As a function of x_n, the expression is a convex quadratic with vertex at $x_n = aP/2$. So on $[x_{n-1}, aP/2]$ it is at most its value at $x_n = x_{n-1}$.

Hint 2. At $x_n = x_{n-1}$ the value is $\leqslant nx_{n-1}^2 - aPx_{n-1} \leqslant (n - a2^{n-2})x_{n-1}^2$, since $P \geqslant 2^{n-2}x_{n-1}$. This is $\leqslant 0$ because $n \leqslant 2^{n-2}$ for $n \geqslant 4$.

Hint 3. Equality forces $a = 1$, $n = 4$ and all $x_i = 2$. Consequence: in a fundamental solution of the Hurwitz equation (one with $x_n \leqslant \frac{a}{2}x_1 \cdots x_{n-1}$), the smallest variable is 1, except for $(2, 2, 2, 2)$ with $a = 1$, $n = 4$.